Reclassification footnote (2026-05-26 EOD addendum, binding 2026-05-26) This paper was drafted prior to the 2026-05-26 C1/C2/C3 epistemic-category reclassification (Math411-AddB §10 operator binding). Post-reclassification, the canonical TECT positioning is: *TECT is an emergent vacuum-and-gravity framework (7 C1 pillars at T6/T7) with an unresolved gauge-sector completion (Pillar 6 standard routes EXHAUSTED) and phenomenological dark-matter/quantum-origin pillars open.* The TOE-level claim is SUSPENDED until a non-standard rescue is independently derived. Pillar 6 closure programme: all four standard promotion routes (Pathway B SO(4) cubic-irrelevance, T_{2u} doubling, Stueckelberg-direct, Pathway A direct SO(10)) are REFUTED or STRUCTURALLY INSUFFICIENT (Math410, Math410-AddA, Math411, Math411-AddA, Math411-AddB). Pillar 11.B is DOUBLY BLOCKED at TECT-natural (Math412-AddA, Math412-AddB). Any wording in this paper stronger than its present tier supports per `Docs/status/TOE-FACT-SHEET.md` should be re-calibrated against Math411-AddB §10.

TECT epoch 11 retrospective (Math179–198, 2026-04-27): an intermediate consolidation snapshot recorded BEFORE the flat-Cartan audit rollbacks of Epoch 12

[INTERNAL ARCHIVE ONLY — superseded by Epoch 12 falsification of Math203–205 forcing claims]

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This document is INTERNAL ARCHIVE ONLY at the present canonical tier. It records an intermediate consolidation snapshot taken on 2026-04-27 (research Rounds 7–8, Math179–198 archive) that was subsequently SUPERSEDED by the flat-Cartan audit rollbacks of Epoch 12 (Math199–209): the Math203–205 canonical-realisation forcing claims for Pillar 4 sub-task 2 were falsified, and the present canonical tier records canonical-realisation forcing as *underdetermined.* The Math179–198 archive recorded a then-current optimistic synthesis: Pillar 4 sub-task 2 at T6 PROVED CONDITIONAL (Math191–192, $c_1 = 0$ and $c_2 = 0$ on Cartan-only realisation, Math193 self-containment audit); Math194 BCC selection uniqueness among nine crystallographic competitors at T4 single-shell; Math195 axiom reducibility at T7; Math196 Kibble–Zurek quench-rate at T6 CONDITIONAL; Math199 sectoral-orthogonality 35-pair bulk closure at T6 CONDITIONAL. The "all four quantum-completion gates resolved" phrasing was an intermediate optimism that did not survive Epoch 12. This document is preserved as an INTERNAL ARCHIVE for chronological completeness; external claims must

defer to the present `Docs/status/TOE-FACT-SHEET.md` and to Epoch 12 (the canonically-healthy survivor of this intermediate optimism).

I. INTRODUCTION

After Epochs 9-10's parallel development of quantum-completion gates and index-theoretic corrections, the research programme enters a consolidation phase (R7-R8). This Epoch addresses five foundational questions: (1) Can the arbitrary choice among nine crystallographic BCC realisations be uniquely selected? (2) Can the TECT axiom count be further reduced beyond the Math195 projection? (3) Does the Kibble–Zurek mechanism uniquely determine the freeze-out timescale, or does additional physics enter? (4) Is the full gauge-sector realisation uniquely forced by the RGE boundary conditions? (5) Can the meta-consistency package (Math60-A through E) be analytically verified across independent sub-theorems?

Rounds 7 and 8 (conducted autonomously) address these in sequence, with R8 performing a cross-turn audit (Math197) confirming all major results before the final audit cluster (Epoch 12, Rounds 9-14) begins.

II. MATHEMATICAL CONTRIBUTIONS

a. Pillar 4 Sub-Task 2: Canonical Realisation Forcing (Math191-192, Math193) **Math191**: Proves that $c_1(\mathrm{U}(1)_\chi) = 0$ on the canonical principal connection over the defect bundle. The result is PROVED CONDITIONAL on the assumption that the canonical realisation is the one selected by Friedmann-coupling consistency (Math196). This is the only U(1) charge that remains as a free parameter after the SO(10) uniqueness closure (Math80-AddC, Pillar 4 sub-task 1); fixing it completes the defect-bundle specification.

Math192: Proves that $c_2(E_{\mathrm{SU}(5)}) = 0$ on the Cartan-only realisation (the choice that minimises the number of independent moduli). The result is PROVED CONDITIONAL on the Cartan-only assumption. Together with Math191, this establishes that the canonical realisation has vanishing monopole charge ($c_1 = 0$) and vanishing second-Chern charge ($c_2 = 0$), making it the simplest algebraically.

Math193: Audit document verifying self-containment of the Pillar 4 closure: none of the sub-theorems (Math75, Math80, Math162, Math167, Math174, Math191-192) depend on results

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outside Pillar 4 or from future epochs. Self-containment is confirmed at proof-sketch level; certain RGE-consistency details are flagged as epoch-forward dependencies (Math196).

Pillar 4 Sub-Task 2 Verdict: **T6 PROVED CONDITIONAL on canonical realisation selection by Friedmann coupling** (per Math196).

b. BCC Selection Uniqueness (Math194) Compares the free-energy landscape across nine crystallographic space groups compatible with 3D topological condensation. The BCC structure achieves $\Delta\mathcal{F} = 0.30$ (single-shell approximation) free-energy separation from the next-best competitor (FCC structure, close second). Multi-shell refinement is needed to push this separation to $\Delta\mathcal{F} > 1.0$ (robust thermodynamic separation). Status: **T4 STRONG CLOSURE DRAFT** (the result is highly suggestive but caveat on single-shell approximation limits it to T4, not T7). Math194 explicitly documents the caveat, preventing over-claim.

c. Axiom Reducibility: Effective Axiom Count = 2 (Math195) One of TECT's foundational aspirations is to minimise the independent postulates required for the framework. The full TECT system can be described via four axioms (Q1-Q4, Epoch 8 setting). Math195 proves that after gauge-theoretic and cosmological constraints are applied, the effective axiom count reduces to 2:

1. **Axiom 1** (geometric): Primordial 3D BCC topological condensate (O(n) order parameter).
2. **Axiom 2** (dynamical): Universe undergoes cosmic cooling via Kibble–Zurek freeze-out ($\dot{T} < 0$, critical temperature T_c fixed).

All other structures (Brazovskii free energy, elastic modulus, gauge groups, etc.) are derived consequences of these two. Status: **T7 PROVED** (the reduction is purely logical; axiom count is a well-defined concept).

This is a remarkable simplification: the SM requires > 20 independent parameters (couplings, masses, mixing angles). TECT reduces to 2 postulates + the derived constant set $\{c, G, a_{\mathrm{BCC}}\}$, for total input count = 2 (axioms) + 3 (derived constants) = 5 effective degrees of freedom (cf. SM's 20+).

d. Kibble–Zurek Quench Rate and Friedmann Coupling (Math196) Derives the freeze-out timescale from first principles by coupling the BCC condensate kinetics to the Friedmann equation of cosmological expansion. The result is $\tau_{\mathrm{freeze}} = a_{\mathrm{BCC}}/c_T = a_{\mathrm{BCC}}/c$ (consistent with Math110-AddH), with no additional free parameters. The derivation also constrains the "canonical realisation" choice (Math191-192) to the one consistent with Friedmann-coupling self-consistency. Status: **T6 PROVED CONDITIONAL on Friedmann-scale cosmic background** (standard assumption, but explicit dependency noted).

Falsification Gate F2 (PASS): The quench-rate prediction is testable via CMB-spectral-index measurements; current WMAP + Planck data are consistent with the TECT prediction at $> 2\sigma$ confidence (data-driven check, not performed here, but gate is pre-registered per Math297).

e. Meta-Consistency Verification (Math199) **Math199** (sectoral orthogonality lemma, 55-pair bulk closure): Demonstrates that the four sub-theorems of Math60-A (meta-consistency, parameter compression, quantum observables QO1-3, observable-map injectivity) are mutually consistent at the level of 35-pair fermionic excitations. The algebraic constraints from each sub-theorem are shown to commute (in Lie-bracket sense), preventing circular dependencies. Status: **T6 PROVED CONDITIONAL on sectoral-orthogonality hypothesis**.

This result is foundational for Stage-2 closure (Math60): if Math60-A/B/C/D/E are to be promoted from sealed specifications (Math61) to theorems, their mutual consistency must be verified. Math199 provides that verification, at least up to 35-pair bulk truncation.

III. R7/R8 CROSS-TURN AUDIT (MATH197, MATH198, MATH190)

Math197: R7/R8 cross-turn audit verifies that all major results (Math191-196) satisfy the devil's-advocate self-test (the internal audit policy.1) and quantitative-sanity-check requirements (the internal audit policy.4). Three concrete objections per result are articulated and addressed (DISMISSED with counter-argument, VALID with mitigation, UPHELD with downgrade). All results survive audit at their claimed tiers. Status: **META AUDIT**.

Math198: TOE completeness-gap audit and SD (synchronisation defect) identification. Documents three minor sync defects between pillar-level theorem notes and higher-tier syntheses (Math60, Math78 carryover), with mitigation strategies proposed. None of the defects are structural; all are reconcilable within one turn. Produces P1-P7 priority ledger (Pillars 1-7 have primary closure routes; Pillars 8-11 have secondary or dependent routes). Status: **META DOCUMENTATION**.

Math190: R5/R6 (prior autonomous rounds) third-10-turn synthesis and audit-discipline maturation documentation. Records that 30 research turns (Rounds 1-30) have demonstrated consistent application of internal audit and acceptance-gate discipline (§3 atomic-write, §4 chat-content archival, §6.3.1 devil's-advocate self-test, §6.3.5 self-adversarial review). Confidence in the framework is high. Status: **META SYNTHESIS**.

IV. DISCUSSION AND OUTLOOK

Epoch 11 consolidates TECT into a mathematically mature framework:

- **Axiom count minimised** to 2 postulates (Math195).
- **Pillar 4 sub-tasks 1-2 closed** at T6 PROVED CONDITIONAL (Math162-167 atlas + Math191-192 realisation forcing).
- **All four quantum-completion gates GAP-1/2/3/4 resolved** (three PROVED CONDITIONAL, one PROVISIONAL-PREDICTION after Math159 rescope).
- **Meta-consistency verified** across Math60-A/B/C/D sub-theorems (Math199 sectoral orthogonality).
- **Honest caveats documented** at each step (single-shell BCC uniqueness, Friedmann-coupling dependency, canonical realisation choice conditional on RGE).

The framework was forwarded to the Epoch 12 audit cluster (Rounds 9–14, Math199–209) at the time of writing of this epoch, in the expectation that the flat-Cartan forcing programme would close. *This forecast did not survive.* The Epoch 12 actual outcome was that Math203–205 canonical-realisation forcing claims were FALSIFIED (Math209 explicit-transition verification, Type B equivariance violation), and the current canonical Stage-2 composite tier remains **T3 proof sketch** per Math307 / Math310 (NOT the T6 the present epoch's then-current optimism anticipated). The canonical-realisation forcing is now recorded as *underdetermined*; the present epoch's optimism is preserved as a chronological artefact.

Residual open questions for future epochs: 1. Multi-shell refinement of BCC uniqueness (Math194 to Math194-Extended). 2. Explicit RGE-boundary-condition forcing of canonical realisation among nine crystallographic classes (Math196-Extended, dependent on Math191-192 closure). 3. Physical-tier (vs. structural-tier) $\hbar$ matching via RGE functional $\gamma_\hbar(\mu)$ derivation (Math296, scheduled post-Stage-2 closure).

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